

The Race between Man and Machine

Tasks, wages, and the limits of formal verification

davidolano03

Acemoglu & Restrepo (2018), American Economic Review

github.com/davidolano03/ai-06-restrepo

AI and Economic Modeling September 2026

Source read: NBER revision of June 2017, 87 pages

Working-paper title: *The Race Between Machine and Man*

The economic question

When machines can perform more tasks, what happens to **wages, employment, and labor's share?**

- ▶ Automation makes existing tasks feasible with capital.
- ▶ New tasks give labor a new productive role.
- ▶ Factor prices influence which tasks are adopted and which innovations pay.

The unit of analysis is an **aggregate economy**. A worker's wage and the economy's labor share need not move together.

NBER 22252, June 2017, pp. 1–6.

Tasks and the adoption threshold

$$i \in [N - 1, N], \quad \gamma'(i) > 0, \quad I \in [N - 1, N].$$

Capital is technologically feasible only for $i \leq I$. Firms compare unit factor costs:

$$R \quad \text{versus} \quad \frac{W}{\gamma(i)}, \quad \frac{W}{R} = \gamma(\tilde{I}), \quad I^* = \min\{I, \tilde{I}\}.$$

- ▶ Capital performs the lower tasks; labor performs the higher tasks.
- ▶ Raising N replaces the lowest task with a more productive one.
- ▶ Task measure stays at one: new tasks change composition and quality.

NBER 22252, June 2017, eqs. (1)–(6), pp. 6–8.

The household and firms

$$\max_{C,L} \frac{(Ce^{-\nu(L)})^{1-\theta} - 1}{1-\theta}, \quad C = RK + WL, \quad \nu'(L) = \frac{W}{C}.$$

At $\theta = 1$, utility is $\ln C - \nu(L)$. Capital is fixed in the static model.

Competitive firms minimize each task's cost and the final producer combines tasks through a CES aggregator. Factor markets clear.

$$L = L^s(\omega), \quad \omega = \frac{W}{RK}, \quad s_L = \frac{WL}{RK + WL} = \frac{\omega L}{1 + \omega L}.$$

Employment responds to the wage relative to capital income, not to the wage alone.

NBER 22252, June 2017, eqs. (4), (8)–(11), pp. 7–10.

Conditions behind the static results

1. Positive, strictly increasing $\gamma(i)$: labor has comparative advantage in higher tasks.
2. $\eta \rightarrow 0$ or $\zeta = 1$: homothetic factor demand, with $\hat{\sigma} = \sigma(1 - \eta) + \zeta\eta > 0$.
3. Capital small enough that $R > W/\gamma(N)$: new tasks are used immediately.

Also: competitive production; the household's regularity and concavity conditions; positive labor-supply elasticity ε_L .

For strict automation effects we need $N - 1 < I^* = I < \tilde{I}$. At $I = \tilde{I}$, left and right derivatives may differ.

NBER 22252, June 2017, Assumptions 1–3; Propositions 1–3; footnote 15.

Displacement and reinstatement

$$G = \int_{I^*}^N \gamma(i)^{\hat{\sigma}-1} di, \quad a = I^* - N + 1,$$
$$\Lambda_I = \frac{\gamma(I^*)^{\hat{\sigma}-1}}{G} + \frac{1}{a} > 0, \quad \Lambda_N = \frac{\gamma(N)^{\hat{\sigma}-1}}{G} + \frac{1}{a} > 0.$$

With K fixed and $I^* = I < \tilde{I}$:

$$\frac{d \ln \omega}{dI} = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0, \quad \frac{d \ln \omega}{dN} = \frac{\Lambda_N}{\hat{\sigma} + \varepsilon_L} > 0.$$

Labor share and employment move with ω . Automation displaces labor; new tasks reinstate labor demand.

NBER 22252, June 2017, Proposition 2, pp. 11–12.

The wage condition

For a marginal increase in I , holding N, K fixed:

$$\underbrace{\frac{d \ln W}{dI}}_{\text{equilibrium wage}} = \underbrace{\frac{\partial \ln Y}{\partial I} \Big|_{K,L}}_{g_I: \text{ productivity}} - \underbrace{(1 - s_L) \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L}}_{D_I: \text{ displacement}}.$$

The wage rises if and only if $g_I > D_I$.

Both effects are positive in magnitude. Labor share can fall while the wage rises. Productivity holds both factors fixed; the wage response allows employment to adjust.

NBER 22252, June 2017, Proposition 3, pp. 12–14.

A reproducible static experiment

$$\eta \rightarrow 0, \quad \sigma = 0.8, \quad \gamma(i) = e^{5i}, \quad B = 1, \quad N = 1, \quad I = 0.4.$$

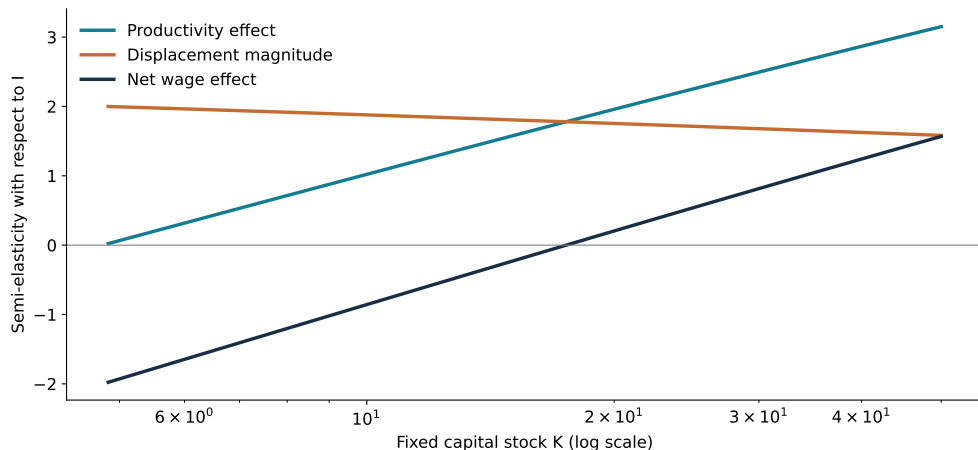
Choose $\nu(L) = L^2/2$ and logarithmic utility. Then

$$L^2 = s_L, \quad \omega = \frac{L}{1 - L^2}, \quad \varepsilon_L = \frac{1 - L^2}{1 + L^2}.$$

- ▶ Solve employment and the adopted cutoff for 220 values of K .
- ▶ Verify market clearing, household optimality and cutoff consistency.
- ▶ Compare the analytical wage response with central differences.

Illustrative parameters, not empirical estimates. All retained points satisfy the new-task adoption condition.

Productivity can dominate displacement



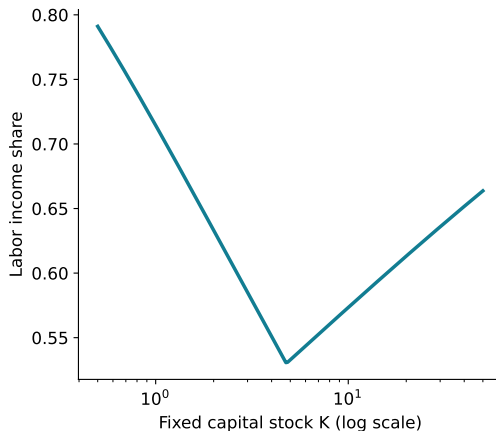
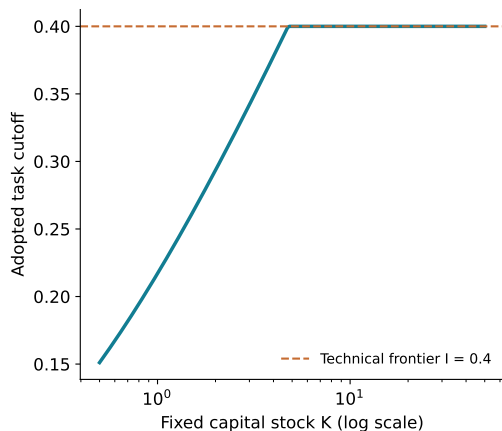
Own computation; technology-constrained points only. Each derivative holds its displayed K fixed.

Two admissible wage responses

	Wage falls	Wage rises
Fixed capital K	8.370	32.834
Productivity g_I	0.777	2.612
Displacement D_I	1.909	1.663
$d \ln W / dI$	-1.132	+0.950
ds_L / dI	-1.375	-1.299
dL / dI	-0.916	-0.811

These are derivatives per unit of I , not percentage changes for a one-unit shock. Both cases satisfy $I^* = I < \tilde{I}$ and $R > W/\gamma(N)$. Maximum analytical/numerical wage-derivative error: 2.3×10^{-9} .

Technological feasibility is not adoption



At $K = 0.5$, $I^* \approx 0.151 < I = 0.4$. A small increase in I has zero local effect. Cross- K slopes are not automation effects.

Balanced growth: a conditional result

Proposition 6 assumes $\gamma(i) = e^{Ai}$, $A > 0$, Assumption 2, $\hat{\sigma} > \zeta$, and sufficiently few scientists: $S < \bar{S}$.

$$\rho > \bar{\rho}, \quad \frac{\kappa_I}{\kappa_N} > \bar{\kappa} \quad \implies \quad n = N - I \in (\bar{n}(\rho), 1), \quad \kappa_I v_I(n) = \kappa_N v_N(n).$$

- ▶ A unique interior balanced growth path balances the two innovations.
- ▶ $\theta = 0$: global saddle-path stability. $\theta > 0$: local uniqueness and asymptotic saddle-path stability.
- ▶ $\rho < \bar{\rho}$: a full-automation BGP exists. Other innovation-productivity ranges allow multiple BGPs or no automation.

Restoring forces after a temporary shock do not undo a permanent change in innovation incentives. Strict threshold equalities are not covered here.

NBER 22252, June 2017, Proposition 6 and Corollary 2, pp. 25–28.

Mathematics, Lean translation, checked proof

Reduced response equations (w, r, ℓ are candidate log changes):

$$\sigma(w - r) + \ell = z, \quad \ell = \varepsilon(w - r), \quad sw + (1 - s)r = g \Rightarrow w = g + (1 - s)\frac{z}{\sigma + \varepsilon}.$$

```
theorem solve_linearized_system
  (s sigma eps g z w r ell : Real) (hden : Not (sigma + eps = 0))
  (hdemand : sigma * (w - r) + ell = z)
  (hsupply : ell = eps * (w - r))
  (haccount : s * w + (1 - s) * r = g) :
  And (w = g + (1 - s) * (z / (sigma + eps)))
    (And (r = g - s * (z / (sigma + eps)))
      (ell = eps * (z / (sigma + eps)))) := by
  have hrel : w - r = z / (sigma + eps) := by
    apply (eq_div_iff hden).2
  nlinarith [hdemand, hsupply]
```

MainTheorems.lean, with ASCII variable names and And/Not notation. Opening proof fragment; full theorem compiles.

What Lean verifies

- ▶ $\mathbb{R}$ fixes the domain. The equations are explicit premises; $\sigma + \varepsilon \neq 0$ makes division valid.
- ▶ `eq_div_iff` converts a quotient identity to multiplication. `nlinarith` proves the polynomial consequence.
- ▶ Setting $z = -\Lambda_I$ recovers the displacement decomposition. Two more contracts construct a reduced-system solution and positive productivity with a negative wage response.

Boundary: these are three algebraic support contracts. Lean does not yet derive the response equations from the continuum economy, prove differentiability, or prove the paper's BGP results.

Lean 4.30.0-rc2: focused proof build **exit 0**; required `-fast` check **exit 0**. Full semantic audit and source theorem coverage remain open.

Hand check: employment and labor share

$$x \hat{\sigma} \quad (\hat{\sigma} + \varepsilon_L) \frac{d \ln \omega}{d I} = - \frac{\gamma(I) \hat{\sigma}^{-1}}{G(I)} - \frac{1}{\alpha}$$

$$\Delta_I = \frac{\gamma(I) \hat{\sigma}^{-1}}{G(I)} + \frac{1}{\alpha} > 0$$

Por tanto: $\frac{d \ln \omega}{d I} = - \frac{\Delta_I}{\hat{\sigma} + \varepsilon_L} < 0$

→ Efecto sobre empleo y part. laboral:

$$\frac{d \ln L}{d I} = \varepsilon_L \frac{d \ln \omega}{d I} < 0.$$

Part. laboral: $s_L = \frac{WL}{rK + WL} = \frac{\omega L}{1 + \omega L}$

Sea $x = \omega L \rightarrow \frac{ds_L}{d I} = \frac{1}{(1+x)^2} \frac{dx}{d I}$

$$\frac{dx}{d I} = x \left(\frac{d \ln \omega}{d I} + \frac{d \ln L}{d I} \right) \Rightarrow \frac{ds_L}{d I} = s_L (1 - s_L) (1 + \varepsilon_L) \frac{d \ln \omega}{d I} < 0$$

Detail from the student's first handwritten page.

With K, N fixed and $I^* = I < \tilde{I}$:

$$\frac{d \ln \omega}{d I} = - \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0.$$

$$\frac{d \ln L}{d I} = \varepsilon_L \frac{d \ln \omega}{d I} < 0.$$

$$\frac{ds_L}{d I} = s_L (1 - s_L) (1 + \varepsilon_L) \frac{d \ln \omega}{d I} < 0.$$

Verdict: displacement lowers employment and labor's share under the stated conditions. This does not determine the wage sign.

Hand check: the wage claim

Handwritten derivation on grid paper:

$$g_I = \frac{d \ln W}{dI} - (1-s_L) \frac{d \ln W}{dI}$$

$$\hookrightarrow \frac{d \ln W}{dI} = g_I - (1-s_L) \frac{\Delta_I}{\hat{\sigma} + \varepsilon_L}$$

$$\hookrightarrow D_I = (1-s_L) \frac{\Delta_I}{\hat{\sigma} + \varepsilon_L} > 0 \quad (\text{magnitude of displacement})$$

$$\rightarrow \frac{d \ln W}{dI} = g_I - D_I \quad \text{para } \hat{\sigma} = 1$$

$$\hookrightarrow g_I = \ln \left(\frac{W}{Rr(I)} \right)$$

Detail from the student's second handwritten page.

Claim to test: "Automation necessarily lowers wages."

$$\frac{d \ln W}{dI} = g_I - D_I,$$

$$D_I = (1-s_L) \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} > 0.$$

Verdict: false in general. The wage rises exactly when $g_I > D_I$.

Productivity holds K, L fixed; the equilibrium wage response allows employment to adjust. An inactive automation frontier has no marginal effect.

What the exercise establishes

1. The sign of wage changes requires comparing productivity with displacement.
2. A lower labor share does not imply a lower real wage.
3. New tasks and endogenous innovation matter for long-run labor demand.
4. A compiling Lean proof verifies its stated premises and conclusion; it does not certify an unproved economic interpretation.

Next research question: which measurable task characteristics predict large automation cost savings, and which predict primarily displacement? The present simulation supplies no causal estimate.

Handwritten derivation: original page 1

Bajo supuestos 1-3: $N-1 < I^w = I < \tilde{I}$

Definimos: $\omega = \frac{w}{RK}$, $\varepsilon_L = \frac{d \ln L}{d \ln \omega} > 0$, $\hat{\sigma} > 0$.

$G(I) = \int_1^N \gamma(i)^{\hat{\sigma}-1} di > 0$, $a = I - N + 1 > 0$

→ Efecto sobre la demanda relativa de trabajo

... (13) $\ln \omega + \frac{1}{\hat{\sigma}} \ln L^S(\omega) = \left(\frac{1}{\hat{\sigma}} - 1 \right) \ln K + \frac{1}{\hat{\sigma}} \ln \left(\frac{G(I)}{a} \right)$

$\frac{d}{dI} \left(\left(1 + \frac{\varepsilon_L}{\hat{\sigma}} \right) \frac{d \ln \omega}{dI} \right) = \frac{1}{\hat{\sigma}} \left(\frac{G'(I)}{G(I)} - \frac{1}{a} \right)$

Por Leibniz: $G'(I) = -\gamma(I)^{\hat{\sigma}-1}$

$\times \hat{\sigma} \quad \left(\hat{\sigma} + \varepsilon_L \right) \frac{d \ln \omega}{dI} = - \frac{\gamma(I)^{\hat{\sigma}-1}}{G(I)} - \frac{1}{a}$

$\Delta_I = \frac{\gamma(I)^{\hat{\sigma}-1}}{G(I)} + \frac{1}{a} > 0$

Por tanto: $\frac{d \ln \omega}{dI} = - \frac{\Delta_I}{\hat{\sigma} + \varepsilon_L} < 0$

→ Efecto sobre empleo y part. laboral:

$\frac{d \ln L}{dI} = \varepsilon_L \frac{d \ln \omega}{dI} < 0$

Part. laboral: $S_L = \frac{WL}{RK + WL} = \frac{\omega L}{1 + \omega L}$

sea $x = \omega L$

$\frac{d S_L}{dI} = \frac{1}{(1+x)^2} \frac{dx}{dI}$

$\frac{dx}{dI} = x \left(\frac{d \ln \omega}{dI} + \frac{d \ln L}{dI} \right) \Rightarrow \frac{d S_L}{dI} = S_L (1 - S_L) \frac{d \ln \omega}{dI} < 0$

Handwritten derivation: original page 2

Productividad y desplazamiento de salario.
 Para $\bar{\epsilon} \neq 1$: $g_I = \frac{\partial \ln Y}{\partial I} \Big|_{K,L} = \frac{\beta \bar{\epsilon}^{-1}}{1-\bar{\epsilon}} \left[\left(\frac{W}{R(I)} \right)^{1-\bar{\epsilon}} - R^{1-\bar{\epsilon}} \right]_{I=0}$

cond de precios $\rightarrow a R^{1-\bar{\epsilon}} + G(I) W^{1-\bar{\epsilon}} = B^{1-\bar{\epsilon}}$

después de dif respecto a I y $\bar{N}$ (constante):

$$0 = R^{1-\bar{\epsilon}} - \bar{\gamma}(I) \bar{\epsilon}^{-1} W^{1-\bar{\epsilon}} + (1-\bar{\epsilon}) \left[a R^{1-\bar{\epsilon}} \frac{d \ln R}{d I} + \right. \\ \left. + G(I) W^{1-\bar{\epsilon}} \frac{d \ln W}{d I} \right]$$

$\hookrightarrow g_I = s_L \frac{d \ln W}{d I} + (1-s_L) \frac{d \ln R}{d I}$

Como K es fijo: $\ln W = \ln W - \ln R - \ln K \rightarrow \frac{d \ln R}{d I} = \frac{d \ln W}{d I} - \frac{d \ln W}{d I}$

$$g_I = \frac{d \ln W}{d I} - (1-s_L) \frac{d \ln W}{d I}$$

$\hookrightarrow \frac{d \ln W}{d I} = g_I - (1-s_L) \Delta_I / (\bar{\epsilon} + \epsilon_L)$

$\hookrightarrow D_I = (1-s_L) \frac{\Delta_I}{\bar{\epsilon} + \epsilon_L} > 0$ (necesidad de desplazamiento)

$\rightarrow \frac{d \ln W}{d I} = g_I - D_I$ Para $\bar{\epsilon} = 1$

$\hookrightarrow g_I = \ln \left(\frac{W}{R \gamma(I)} \right)$